

Heterogeneous LLM-Based Trading Agents in Continuous Double-Auction Markets: A Controlled Evaluation Against Adaptive Benchmarks in ABIDES

Research Proposal

Nguyen Quy Tu, Nguyen Minh Nghia

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Abstract

Continuous double auctions (CDAs) are the dominant trading mechanism in real equity and commodity markets, and a growing body of work shows that large language models (LLMs) can act as plausible trading agents within them—following strategy instructions, emulating archetypes such as value investors and momentum traders, and reproducing emergent phenomena including price discovery and bubbles. This proposal sets out a controlled, single-environment study that evaluates a locally hosted open-source LLM (Gemma 3 4B Instruct) as a trading agent against an adaptive algorithmic benchmark—the Heuristic Belief Learning (HBL) trader native to ABIDES—inside a high-fidelity discrete-event simulator. Across eight experiments, we isolate the effect of prompt structure, memory, information access, decision latency, regime shocks, and inter-agent communication. Evaluation is deliberately focused on three interpretable metrics: bid-ask spread (agent adaptation), price-versus-equilibrium (convergence to fundamental value), and profit-and-loss (whether the LLM actually out-earns the benchmark). We also report preliminary results from the first two experiments—an HBL-based traditional baseline and an LLM-inclusive market—which establish that the environment produces continuous trading and sensible spreads, while surfacing an early, testable observation: the LLM-inclusive market is markedly more active with wider, spikier spreads, yet in both markets the mid-price tracks fundamental value only loosely.

1 Background and Motivation

Continuous double auctions, in which buyers and sellers continuously submit orders to a central limit order book, underpin real-world equity and commodity exchanges. A range of open-source simulators support CDA research, from the public-domain Bristol Stock Exchange (BSE) through to higher-fidelity, discrete-event platforms such as ABIDES [2] and PyMarketSim [7]. These platforms make it tractable to study market microstructure under controlled, repeatable conditions that would be impossible to obtain with human participants.

Recent literature establishes the viability of LLMs as trading agents within such simulations. Lopez-Lira (2025) [6] demonstrates that LLM agents follow strategy instructions consistently—credibly occupying roles such as value investor, momentum trader, and market maker—and that markets populated by such agents reproduce features of real markets including price discovery, bubbles, underreaction, and strategic liquidity provision. The same work warns of a systemic-risk channel specific to LLMs: because agents are highly sensitive to prompts, similar models given similar instructions can act in a correlated manner, amplifying volatility. In the multi-agent setting, Agrawal et al. (2025) [1] study LLM sellers in a CDA and find that the ability to communicate directly increases collusive tendencies, that the propensity to collude varies across models, and that environmental pressures (oversight, urgency) shift collusive behavior—results that directly motivate a dedicated communication-and-collusion experiment.

Despite these advances, important questions remain underexplored. Most published work centers on large, proprietary, API-hosted models; the behavior of **small, locally hosted open-source models** under tight resource and latency budgets is far less understood. The interaction between **memory design, information access, and decision latency** has not been systematically ablated within a single high-fidelity environment, and the **causal effect of LLM agents on aggregate market quality** is rarely isolated against an *adaptive* (rather than zero-intelligence) benchmark. This proposal addresses these gaps directly.

2 Simulator Selection: BSE versus ABIDES

The choice of simulator is consequential, because two of the eight experiments depend on capabilities that not all CDA simulators provide. We compared the Bristol Stock Exchange (BSE; Cliff, 2018 [3]) against ABIDES [2].

Table 1: Comparison of BSE and ABIDES Simulation Environments

Dimension	Bristol Stock Exchange (BSE)	ABIDES
Architecture	Single Python script (~2,500 lines); orders processed sequentially, order book re-published to all traders	Discrete-event, message-passing engine modeled on NASDAQ ITCH/OUCH protocols; nanosecond time resolution
Latency model	None—no exchange-processing delay, no network latency, no simultaneously issued orders	Configurable pairwise agent–exchange network latency plus per-agent computation delays
Communication	None—agents observe only the re-published order book	All agents communicate solely via standardized messages; supports private messaging between agents
Adaptive agents	GVWY, SHVR, ZIC, ZIP, AA, GDX	Zero-Intelligence (ZI) and Heuristic Belief Learning (HBL); plus value, momentum, market-maker, and noise agents
Value process	Exogenous customer limit-price schedules	Dynamic, mean-reverting common-value (fundamental) process with dividends
Scale	Lightweight; embarrassingly parallel across many runs	Tens of thousands of agents per simulation
Best suited to	Teaching, rapid strategy sweeps, prototyping	High-fidelity microstructure, latency and communication studies, RL via ABIDES-Gym

Why ABIDES is required for this study. BSE is an excellent, transparent, and fast platform, and we may use it for quick agent sanity checks. But its synchronous, latency-free design is disqualifying for the core experiments:

1. **Latency sensitivity (Exp 6) needs a latency model.** BSE processes orders sequentially and explicitly assumes no exchange-processing or network delay; it therefore cannot represent the reaction-speed disadvantage an LLM incurs from slow inference. ABIDES models pairwise agent–exchange latency and agent computation delays directly.
2. **Communication and collusion (Exp 8) needs message-based private communication.** In BSE, agents only observe the re-published book and cannot exchange messages; ABIDES routes everything through a standardized message protocol and supports private inter-agent messaging.
3. **An adaptive, order-book-aware benchmark ships natively.** HBL is built into ABIDES; the

GDX/AA strategies are BSE-native and not part of ABIDES, so HBL is the natural in-platform adaptive benchmark.

4. A **dynamic fundamental value process** enables value investors, the price-versus-equilibrium analysis, and the mid-session regime shock (Exp 7).
5. **ABIDES-Gym** provides a convenience wrapper for structured, repeated, reproducible runs.

To keep results comparable, the study uses ABIDES exclusively for all reported experiments, with a constant market size of **exactly 10 agents** per run.

3 Agent Design and Architecture

LLM integration. Agents are powered by a single locally hosted open-source model, **Gemma 3 (4B) Instruct**, removing API cost and network variance and enabling tight control over inference latency. Running a small model also makes the latency-sensitivity experiment (Exp 6) realistic rather than artificial.

Background / benchmark agents. The primary background and benchmark trader is the **Heuristic Belief Learning (HBL) agent** native to ABIDES (after Gjerstad & Dickhaut, 1998 [4]). HBL forms beliefs about the probability that a candidate bid or ask will transact by exploiting recent order-book history, then quotes to maximize expected surplus. Because it is order-book-aware and adapts its quotes to observed market state, HBL is a far more demanding and realistic benchmark than zero-intelligence trading, and it is the appropriate native comparator in ABIDES. A small number of **Zero-Intelligence (ZI)** agents [5] provide baseline liquidity and noise.

Role specification (LLM). System prompts define explicit archetypes: value investors using a dividend-discount valuation, momentum traders keyed to recent returns, and market makers quoting two-sided around a reference price.

Structured outputs. Each LLM agent emits a two-tier output: natural-language reasoning for qualitative inspection, plus a machine-readable JSON order specifying action, quantity, and price. Orders are parsed automatically with a deterministic fallback (e.g., no-op) on malformed output.

Memory buffers. LLM agents use a memory module to store past outcomes—for example, previous fill prices and realized PnL—so that relevant context can be injected into subsequent prompts. Memory horizon is itself a treatment variable (Exp 4).

4 Experimental Methodology

All experiments run locally in Python using ABIDES-Gym as a wrapper for structured, repeated runs. Each run uses exactly 10 agents. To support inference, **each condition is repeated across multiple random seeds** (target: ≥ 10 seeds per condition), with results reported as means and dispersion rather than single runs. The background population is built from HBL agents plus a small number of ZI liquidity providers/takers.

- **Experiment 1 — Baseline Market Reproduction.** Ten background agents (HBL plus ZI) to verify the environment yields sensible bid-ask spreads, convergence toward fundamental value, and continuous volume.
- **Experiment 2 — LLM in the Market vs. Background.** Insert LLM trader(s) into the HBL/ZI background; compare spread behavior, price-versus-equilibrium tracking, and PnL against the benchmark.
- **Experiment 3 — Prompt Ablation.** Vary the LLM prompt across minimal, reasoning, and structured-JSON formats to test whether explicit reasoning improves behavior or merely increases latency.

- **Experiment 4 — Memory Ablation.** Restrict the single LLM’s history across four levels—none, last 5 events, last 20 events, and a rolling session summary—to test adaptation.
- **Experiment 5 — Information-Access Ablation.** Vary the market view across top-of-book only; full visible LOB; LOB plus recent trades; and LOB plus inventory/recent PnL, to isolate the effect of information richness.
- **Experiment 6 — Latency Sensitivity.** Add artificial decision delays (none, small, moderate, large) via the ABIDES latency model to evaluate competitiveness when reaction-speed advantages are lost.
- **Experiment 7 — Shock-Response Market.** Introduce a sudden change in fundamental value or supply–demand regime mid-session; measure adaptation speed, profit loss during the shock, and how quickly mid-price re-converges to the new equilibrium.
- **Experiment 8 — Communication and Collusion.** Shift composition to 2 LLM traders and 8 background agents, placing both LLMs on the same side of the market. Vary private messaging before quoting across three conditions—no communication, restricted communication, free communication—to test inter-agent coordination, observed through spread widening and seller-side PnL.

5 Evaluation Metrics

Evaluation is deliberately focused on **three interpretable metrics**. Each is chosen because it answers a specific question about agent adaptation, market efficiency, or LLM competitiveness; richer microstructure statistics are left out to keep the analysis legible and the comparison clean.

1. **Bid-ask spread—*how agents adapt*.** The spread is the gap between the best bid and best ask. A wide spread means buyers and sellers disagree about price and have not yet adjusted toward one another; a narrowing spread is direct evidence that agents are adapting their quotes in response to the book. Tracking the spread over time therefore reveals the *adaptation dynamics* of each agent population and whether the LLM tightens or widens the market relative to the HBL benchmark.
2. **Price versus equilibrium—*convergence and pattern discovery*.** We compare the realized mid-price (and transaction prices) against the equilibrium/fundamental-value reference that ABIDES generates. This shows whether agents discover the underlying equilibrium pattern and whether the market, on average, converges or reverts toward it [8]. It measures *price efficiency*: does the LLM help the market find fair value faster, track it more closely, or instead push prices away from it?
3. **Profit-and-loss (PnL)—*is the LLM actually better?*** Per-agent realized and mark-to-market PnL answers the bottom-line question directly: does the LLM trader earn more than the HBL benchmark under the same conditions? PnL is the arbiter of competitiveness across every experiment, and in Exp 8 the seller-side PnL/surplus also serves as the collusion signal (do colluding LLMs extract abnormal surplus?).

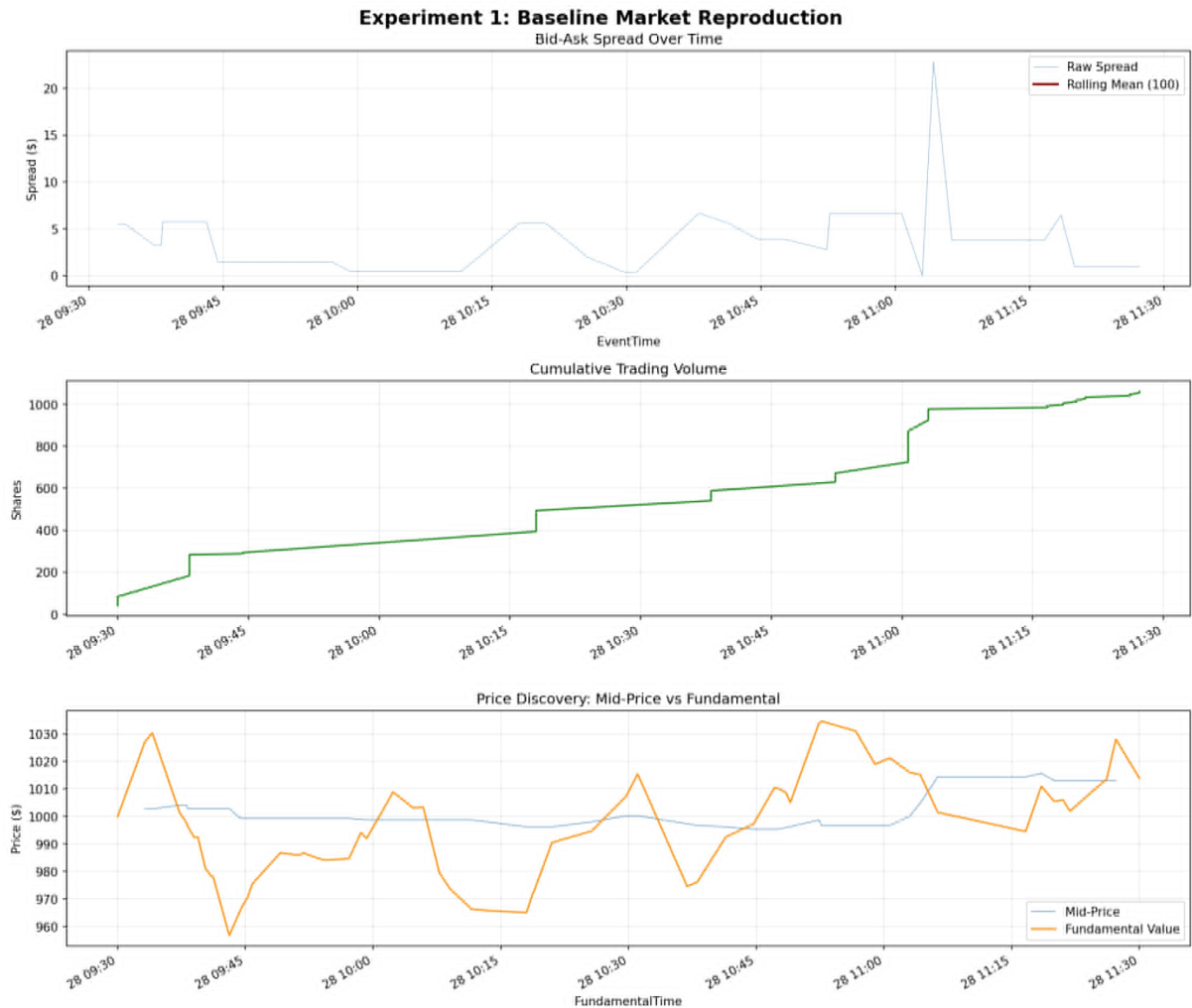
Supporting qualitative check: LLM reasoning traces are inspected manually to confirm strategy adherence and to characterize failure modes (hallucinated prices, malformed JSON, strategy drift). This is a sanity check on the three quantitative metrics, not a separate scoring axis.

6 Preliminary Results (Experiments 1 & 2)

Two pilot runs have been completed over a roughly two-hour session ($\approx$ 09:30–11:30): an HBL/ZI background baseline (Exp 1) and an LLM-inclusive market (Exp 2). *These are single-run pilots without seed replication or significance testing; the figures are read qualitatively and treated as proof-of-concept, not confirmed effects. PnL is not yet plotted, so the pilots speak only to the spread and price-versus-equilibrium metrics.*

6.1 Experiment 1 — HBL/ZI Baseline

- **Spread.** Raw bid-ask spread sits mostly in the **\$0–7** range with a single transient spike to $\approx \$23$. The 100-observation rolling mean is effectively absent, consistent with **fewer than 100 spread observations** over the session—i.e., a relatively sparse, calm event stream. Spreads are tight and well-behaved.
- **Volume (descriptive).** Cumulative volume rises in a stepped fashion to roughly **1,060 shares**, with flat no-trade plateaus between bursts.
- **Price versus equilibrium.** The mid-price is **sticky**, holding a narrow \$995–1015 band, while the fundamental value swings widely ($\approx \$958$ –1034). Mid-price tracks the equilibrium reference only loosely.

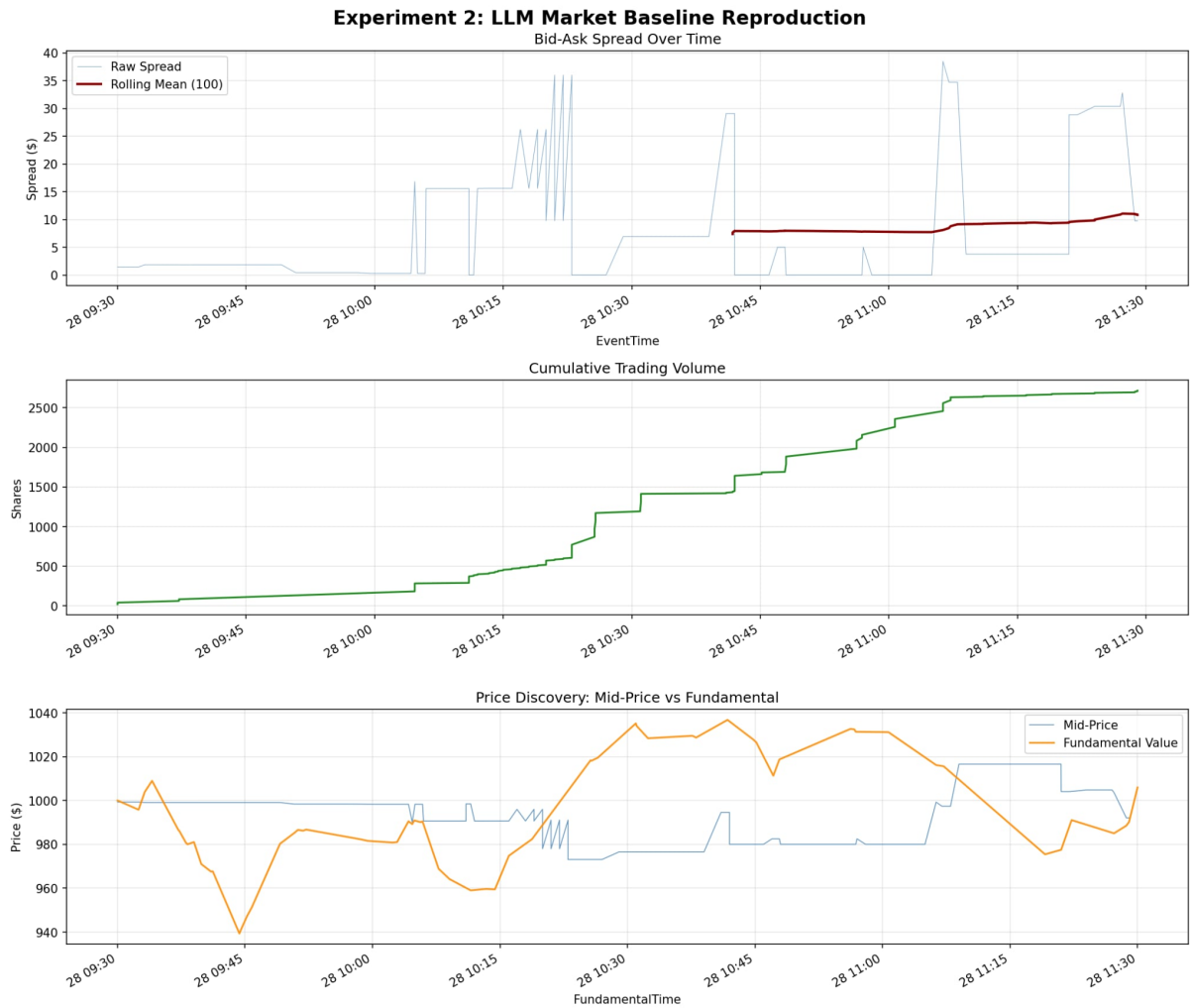


Read. The baseline is calm and orderly—tight spreads, modest but continuous volume—but price discovery relative to a fast-moving fundamental is weak. This is itself useful: it suggests the HBL/ZI mix is not strongly fundamental-aware on this timescale, or that the fundamental process moves faster than the agents re-center. Quantifying the tracking gap (metric 2) is the immediate next step.

6.2 Experiment 2 — LLM-Inclusive Market

- **Spread.** Spreads are visibly **spikier**: near-zero early, then frequent excursions to **\$15–36** through the middle of the session, with a 100-observation rolling mean of about **\$8 rising to $\approx \$11$** . The LLM-inclusive market is both more active and more dispersed in spread than the HBL baseline.
- **Volume (descriptive).** Cumulative volume climbs smoothly to roughly **2,700 shares—about $2.5\times$ the baseline.**

- **Price versus equilibrium.** Mid-price remains sticky ($\approx \$973\text{--}1018$, moving in discrete steps) while the fundamental ranges $\$940\text{--}1037$; tracking is again loose, with extended $\$30\text{--}40$ divergences.



Read. Introducing the LLM substantially raises activity (volume, event rate) and widens/roughens the spread, but it does **not** visibly improve convergence toward fundamental value. The headline early finding is therefore a trade-off: more liquidity and trading, but no gain—possibly a loss—in price efficiency. Whether the LLM also out-earns HBL is the open PnL question the full study must answer.

6.3 Caveats and immediate next steps

1. **Composition of Exp 2.** A $2.5\times$ jump in volume is large; confirm whether the pilot used a single LLM among nine background agents or a more LLM-heavy “LLM market”. This determines how the result is interpreted.
2. **Statistical weight.** With one run per condition, the spread and tracking differences are suggestive, not established. First task: seed replication (≥ 10 seeds) with means and confidence intervals.
3. **Add the PnL panel.** The pilots plot spread and price-versus-fundamental but not PnL; the third core metric needs to be logged and charted before competitiveness claims can be made.
4. **Quantify price-versus-equilibrium.** Replace the qualitative “stickiness” description with a tracking-error number (e.g., RMSE of mid-price vs. fundamental) so the metric is comparable across all eight experiments.

7 Expected Contributions

1. A reproducible, single-environment benchmark for a **small, locally hosted open-source LLM** as a CDA trading agent, evaluated against an **adaptive (HBL) benchmark** rather than zero-intelligence baselines.
2. A factor-by-factor account of which **agent-design choices** (prompt, memory, information, latency) drive spread behavior, price efficiency, and PnL, with explicit latency/quality trade-offs.
3. Evidence on whether LLM agents **improve or degrade price efficiency** while changing liquidity, relative to an adaptive benchmark.
4. A controlled test of **communication-induced coordination/collusion** between same-side LLM agents, measured through spread widening and seller-side PnL.

8 Risks and Limitations

- **Model capacity.** A 4B model may produce malformed outputs or weak reasoning; the JSON-with-fallback design and manual reasoning checks mitigate and surface this.
- **Latency confound.** Local inference latency interacts with the event-driven engine; Exp 6 is designed to characterize rather than eliminate this, which is precisely why ABIDES’s latency model is required.
- **External validity.** Simulated CDAs abstract away many real-market features; claims are scoped to the environment.
- **Single-model scope.** Using one model isolates design effects but limits cross-model generalization; multi-model replication is noted as future work.

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